CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 19: Constructions (Circles)

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Circumscribed circle of a triangle

1. Using ruler and compasses only, construct $\triangle ABC$ with $AB = 6$ cm, $AC = 4.5$ cm and $\angle BAC = 120°$. Construct the circle that passes through A, B and C, and measure its radius. ICSE 2025, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of $120°$ (two $60°$ arcs) and cut off $AC = 4.5$ cm on its arm. Join BC.

Draw the perpendicular bisectors of AB and AC. Let them meet at O.

With centre O and radius OA, draw the circle. It passes through A, B and C.

Because $\angle A$ is obtuse, O lies outside the triangle, on the other side of BC from A.

Check by calculation: $BC^2 = 6^2 + 4.5^2 - 2(6)(4.5)\cos 120° = 83.25$, so $BC \approx 9.12$ cm and the radius is $\dfrac{BC}{2\sin 120°} \approx 5.27$ cm.

Answer: the radius measures about 5.3 cm

2. Using ruler and compasses, construct $\triangle ABC$ with $AB = 6$ cm, $\angle BAC = 120°$ and $AC = 5$ cm. Construct the circle passing through A, B and C, and measure its radius. ICSE 2023, 4 marks

SOLUTION

Draw $AB = 6$ cm. At A construct an angle of $120°$ and cut off $AC = 5$ cm. Join BC.

Draw the perpendicular bisectors of AB and AC, meeting at O. With centre O and radius OA, draw the circle.

Check by calculation: $BC^2 = 6^2 + 5^2 - 2(6)(5)\cos 120° = 91$, so $BC \approx 9.54$ cm and the radius is $\dfrac{BC}{2\sin 120°} \approx 5.5$ cm.

Answer: the radius measures about 5.5 cm

MathSetu

3. Using ruler and compasses, construct $\triangle ABC$ with $AB = 3$ cm, $BC = 4$ cm and $\angle ABC = 90°$. Construct its circumscribing circle and measure the radius. ICSE 2020, 4 marks

SOLUTION

Draw $BC = 4$ cm, construct a right angle at B and cut off $BA = 3$ cm. Join AC.

Draw the perpendicular bisectors of AB and BC, meeting at O. With centre O and radius OA, draw the circle.

Check: in a right-angled triangle the circumcentre is the mid-point of the hypotenuse. $AC = \sqrt{3^2 + 4^2} = 5$ cm, so the radius is 2.5 cm.

Answer: radius $= 2.5$ cm

4. Using ruler and compasses only, construct $\triangle ABC$ with $BC = 5$ cm, $AB = 6.5$ cm and $\angle ABC = 120°$. (a) Construct its circumcircle. (b) Construct the cyclic quadrilateral $ABCD$ in which D is equidistant from AB and BC. ICSE 2018, 4 marks

SOLUTION

Draw $BC = 5$ cm. At B construct an angle of $120°$ and cut off $BA = 6.5$ cm. Join AC.

(a) Draw the perpendicular bisectors of AB and BC, meeting at O. With centre O and radius OB, draw the circle through A, B and C.

(b) Points equidistant from AB and BC lie on the bisector of $\angle ABC$. Draw this bisector and let it meet the circle again at D. Join AD and CD.

Check: the bisector cuts the arc AC in half, so $AD = CD$, and $\angle ADC = 180° - 120° = 60°$. So $\triangle ADC$ should be equilateral.

Answer: $ABCD$ is the required cyclic quadrilateral, with $AD = CD$

2 Tangents from an external point

5. Using ruler and compasses, draw a circle of radius 4 cm with centre O. Mark a point P 7 cm from O and construct the two tangents from P to the circle. Measure the length of a tangent. ICSE 2019, 4 marks

SOLUTION

Draw the circle and mark P with $OP = 7$ cm.

Bisect OP at M. With centre M and radius MO, draw a circle cutting the given circle at A and B.

Join PA and PB. These are the tangents, because $\angle OAP = 90°$ (angle in a semicircle).

Check: $PA = \sqrt{7^2 - 4^2} = \sqrt{33} \approx 5.74$ cm.

Answer: each tangent measures about 5.7 cm

3 Inscribed circle of a triangle

6. Using ruler and compasses, construct $\triangle ABC$ with $AB = 7$ cm, $\angle CAB = 60^\circ$ and $AC = 5$ cm. Construct the locus of points equidistant from (a) AB and AC (b) BA and BC. Hence construct the circle that touches the three sides of the triangle internally. ICSE 2017, 4 marks

SOLUTION

Draw $AB = 7$ cm, construct $\angle CAB = 60^\circ$ and cut off $AC = 5$ cm. Join BC.

(a) The locus is the bisector of $\angle A$. (b) The locus is the bisector of $\angle B$.

The bisectors meet at I, the incentre. From I draw $IN \perp AB$. With centre I and radius IN, draw the circle. It touches all three sides.

Check: $BC = \sqrt{49 + 25 - 35} = \sqrt{39} \approx 6.24$ cm, the area is $\frac{1}{2} \times 7 \times 5 \times \sin 60^\circ \approx 15.16$ cm² and the semi-perimeter is about 9.12 cm, so the radius is about $\frac{15.16}{9.12} \approx 1.7$ cm.

Answer: the incircle has a radius of about 1.7 cm

These questions are based on questions asked in past board examinations. They have been reworded by MathSetu, the figures are redrawn, and the solutions are MathSetu's own. MathSetu is an independent study resource and is not affiliated with or endorsed by CBSE, CISCE, NCERT or any publisher.